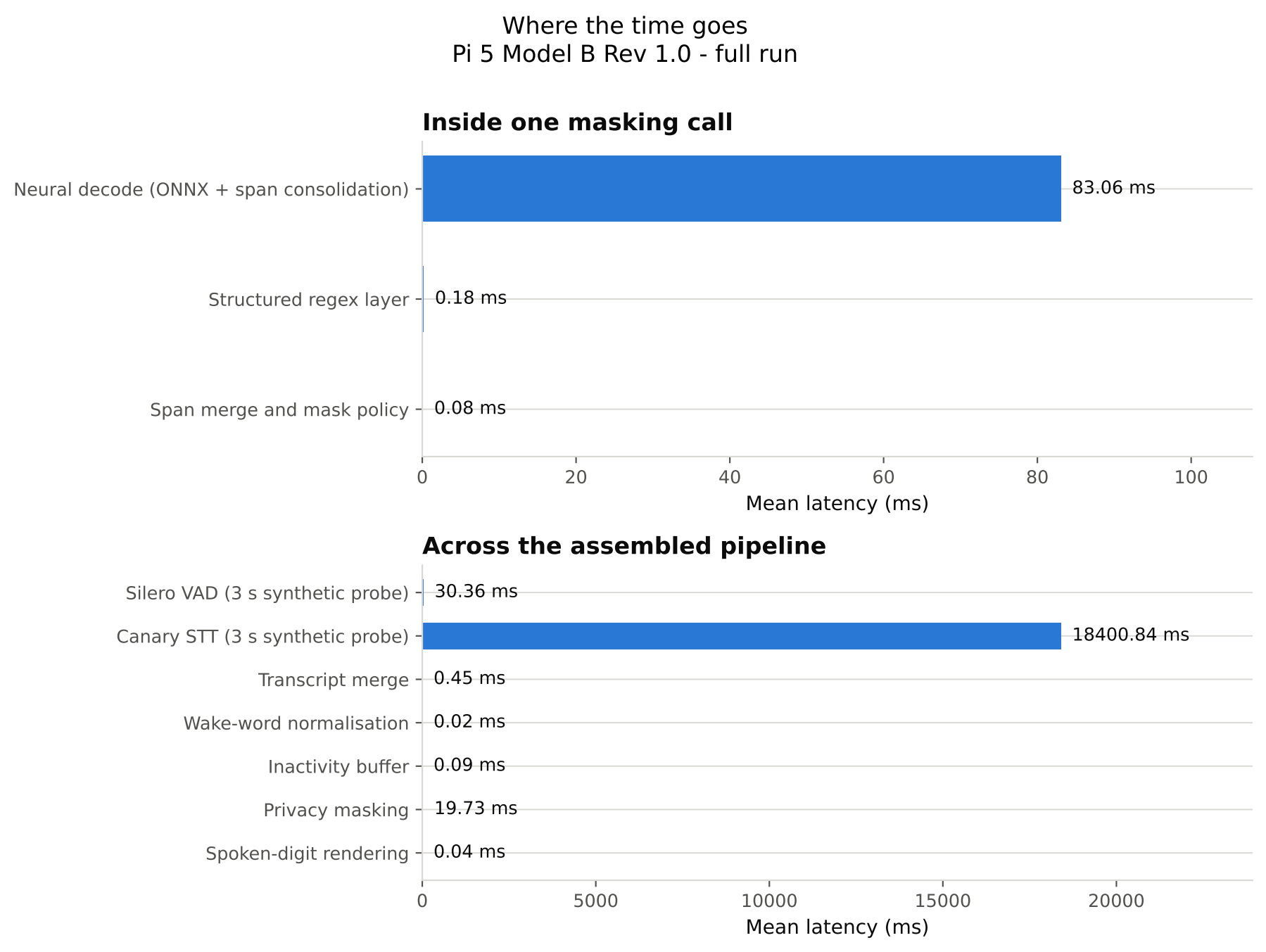
Where the time goes
Pi 5 Model B Rev 1.0 - full run
Inside one masking call
Neural decode (ONNX + span consolidation)
83.06 ms
Structured regex layer
0.18 ms
Span merge and mask policy
0.08 ms
0
20
40
60
80
100
Mean latency (ms)
Across the assembled pipeline
Silero VAD (3 s synthetic probe)
30.36 ms
Canary STT (3 s synthetic probe)
18400.84 ms
Transcript merge
0.45 ms
Wake-word normalisation
0.02 ms
Inactivity buffer
0.09 ms
Privacy masking
19.73 ms
Spoken-digit rendering
0.04 ms
0
5000
10000
15000
20000
Mean latency (ms)